

Microsurgical Tubular Resection of Intradural Extramedullary Spinal Tumors With 3-Dimensional-Navigated Localization

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BACKGROUND: The safety and efficacy of minimally invasive spine surgical (MISS) approaches have stimulated interest in adapting MISS principles for more complex pathology including intradural extramedullary (IDEM) tumors. No study has characterized a repeatable approach integrating the MISS surgical technique and 3-dimensional intraoperative navigated localization for the treatment of IDEM tumors.

OBJECTIVE: To describe a safe and reproducible technical guide for the navigated MISS technique for the treatment of benign intradural and extradural spinal tumors.

METHODS: Retrospective review of prospectively collected data on 20 patients who underwent navigated microsurgical tubular resection of intradural extramedullary tumors over a 5-year period. We review our approach to patient selection and report demographic and outcomes data for the cohort.

RESULTS: Our experience demonstrates technical feasibility and safety with a 100% rate of gross total resection with no patients demonstrating recurrence during an average follow-up of 20.2 months and no instances of perioperative complications. We demonstrate favorable outcomes regarding blood loss, operative duration, and hospital length of stay.

CONCLUSION: Navigated localization and microsurgical tubular resection of IDEM tumors is safe and effective. Adherence to MISS principles and thoughtful patient selection facilitate successful management of these patients.

KEY WORDS: Minimally invasive spine surgery, Tubular retractor, Intradural extramedullary, Spinal tumor, Intraoperative navigation, Total navigation

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The successful treatment of symptomatic intradural extramedullary (IDEM) spinal tumors has traditionally been achieved using an open laminectomy and wide exposure of the spinal canal. Open approaches for resection of IDEM tumors have been shown to be safe and effective in achieving high rates of gross total resection (GTR).^{1–10} However, they can result in significant postoperative pain and iatrogenic instability.^{11–15}

Previous studies on MISS resection of IDEM tumors have been restricted to case reports or small series.^{5,6,16–24} Furthermore, there has been no published description of a practical workflow

integrating intraoperative navigated localization (IONL) with microsurgical tubular resection for the treatment of this population.

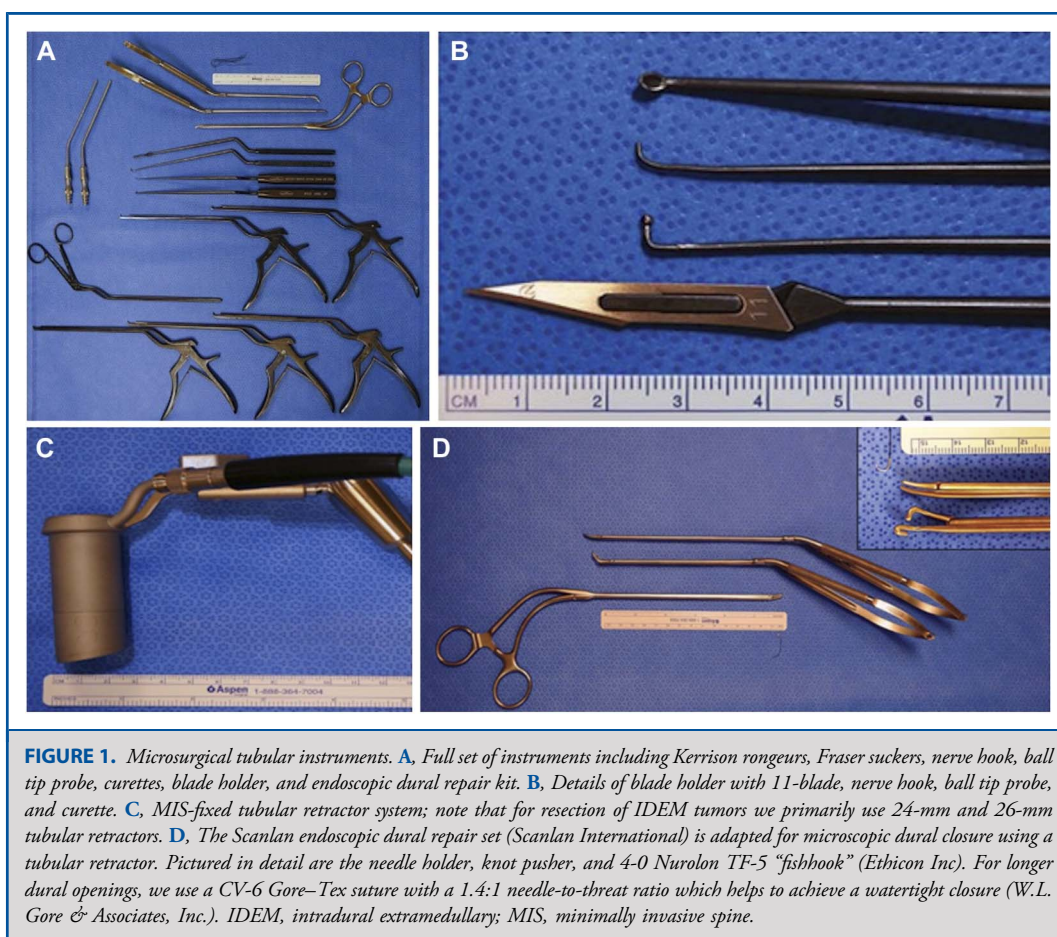
Here, we report our experience implementing a systematic MISS approach to the treatment of benign tumors and a detailed step-by-step description using both total IONL and microsurgical tubular resection.

METHODS

We performed a retrospective review of prospectively collected data on 20 consecutive patients who underwent navigated microsurgical tubular resection of IDEM spinal tumors at our department over a 5-year period between January 2015 and December 2020. Informed consent for surgery was obtained from all patients after review of risks, benefits, and alternatives of the proposed surgery.

The study was reviewed and approved by the Institutional Review Board. The participants and any identifiable individuals consented to publication of his/her image. The

ABBREVIATIONS: GTR, gross total resection; ICT, intraoperative computed tomography; IDEM, intradural extramedullary; IONL, intraoperative navigated localization; MISS, minimally invasive spine surgical; ULBD, unilateral laminotomy for bilateral decompression.



inclusion criteria were cases of primary benign IDEM spinal tumors. All patients underwent surgical resection using the same tubular retractors, intraoperative computed tomography (iCT) scanner (AIRO, Mobius Imaging), and 3-dimensional IONL (Brainlab AG).

In addition to a review of our data, we synthesized our MISS operative experience as previously published to describe a repeatable surgical workflow specifically for IDEM tumors.²⁵⁻³⁵

Indications

Patients with benign tumors of the spine that are symptomatic or that demonstrate growth are generally considered to represent good candidates for surgical resection.^{4,10,36,37}

Tumor size and location are the primary factors that determine the suitability of the MISS tubular approach. We find 24-mm to 26-mm fixed tubular retractors optimal for the resection of tumors that can be

TABLE 1. Ten Steps for 3-Dimensional-Navigated Microsurgical Tubular Resection of IDEM Tumors

1. iCT with navigated incision planning
2. Dilation and docking of tubular retractor on inferior edge of the overlying lamina
3. Laminotomy including over-the-top decompression with additional drilling of the superior margin of the inferior lamina as judged by tumor localization using ION
4. Dural opening and placement of tack up sutures
5. Identification and separation of neural tissue (spinal cord and nerve roots) from the mass
6. Isolation of the mass and stimulation of any entering/exiting nerve roots
7. Sectioning of nonfunctional neural tissue or sparing of functional neural tissue in cases in which the mass is too adherent or involved to permit GTR
8. Resection of the mass
9. Watertight dural closure with Scanlan endoscopic dural repair instruments and CV-6 Gore-Tex suture followed by Valsalva maneuver to ensure watertight seal
10. Multilayered closure to minimize risk of pseudomeningocele formation

GTR, gross total resection; iCT, intraoperative computed tomography; IDEM, intradural extramedullary; ION, intraoperative navigated.

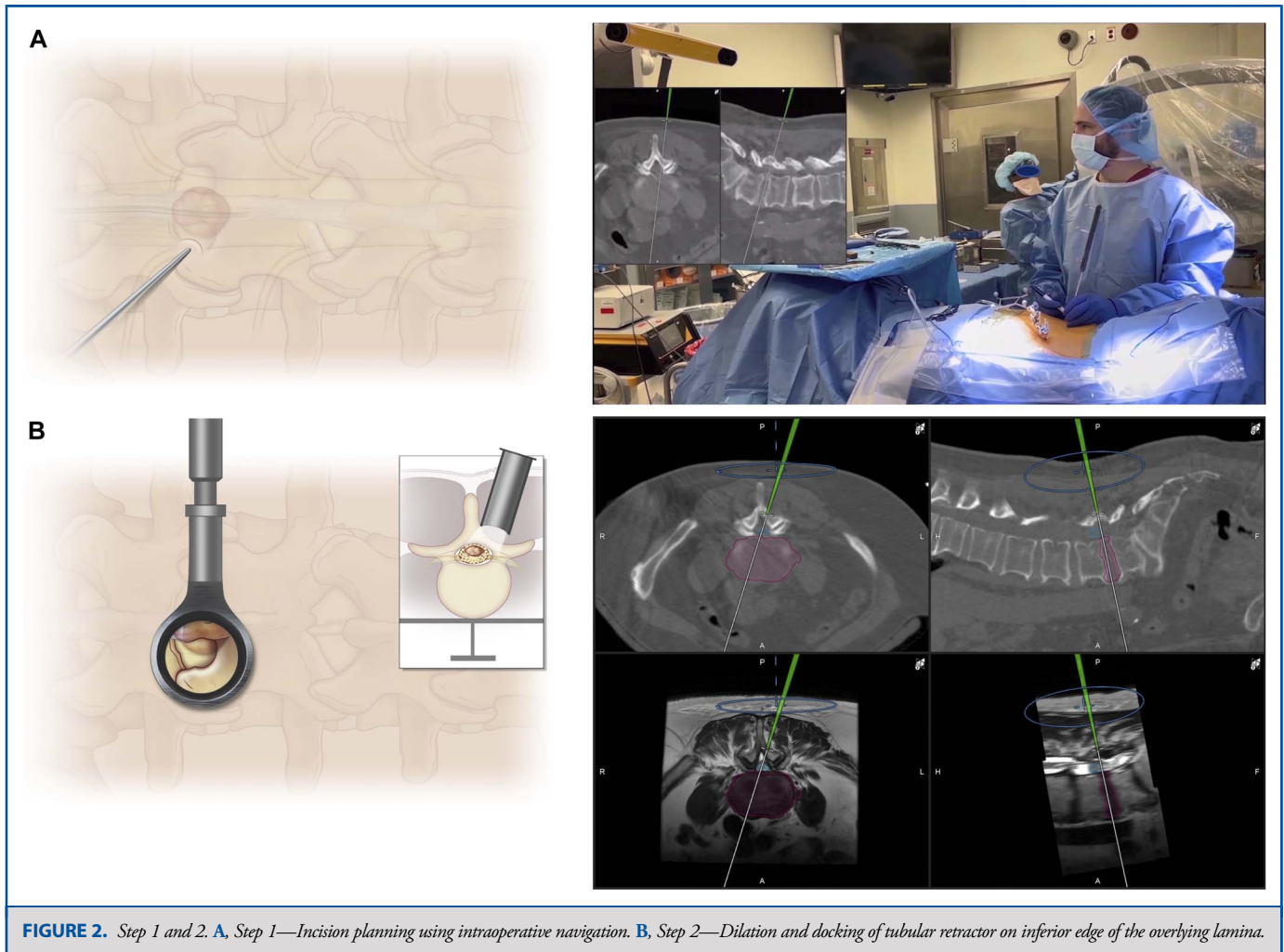


FIGURE 2. Step 1 and 2. **A,** Step 1—Incision planning using intraoperative navigation. **B,** Step 2—Dilation and docking of tubular retractor on inferior edge of the overlying lamina.

accessed with a 1 to 2 level laminotomy. “Wandering” of a tubular retractor facilitates resection of tumors up to 2 laminae in maximum dimension.

Tumors that arise below the conus medullaris can typically be approached dorsally regardless of their position within the thecal sac. However, tumors arising above the conus medullaris must be considered in terms of the degree of spinal cord manipulation required. Ventral tumors are safer to approach using open techniques.

Instrumentation

Appropriate surgical instrumentation is a critical component of MISS tubular resection of IDEM tumors. Our preferred instrumentation is outlined in Figure 1. Intraoperative neuromonitoring is used in all cases. For IDEM tumors above the conus, we use motor-evoked potentials, somatosensory-evoked potentials, and triggered electromyography; for IDEM tumors below the conus medullaris, only triggered electromyography is used.

General Technique

We approach selected IDEM tumors using unilateral laminotomy through a tubular retractor. Our technique for performing unilateral

laminotomy for bilateral decompression (ULBD) has been described previously.^{25,38} We use these same MISS techniques for the approach to the tumor and dural closure. The microsurgical principles that guide dural opening and microsurgical resection of the tumor remain constant across open and MISS approaches (Table 1).

Localization and Incision

We use the “total navigation” approach acquiring an iCT scan and registering our intraoperative navigated (ION) platform at the outset of the procedure^{39,40} (Figure 2A). A reference array is secured into the posterior iliac crest. After ION registration, we use the navigated probe for incision planning.⁴¹ An approximately 3-cm linear skin incision is made 1 to 2 cm off midline, on the side to which the tumor is eccentric. The fascia is incised sharply using a 15-blade.

Tubular Retractor Placement and Exposure of Bony Anatomy

The first dilator is passed through the musculature onto the lamina. The dilators are sequentially enlarged until the desired diameter (typically 24–26 mm) is reached, and the appropriate length tubular retractor is

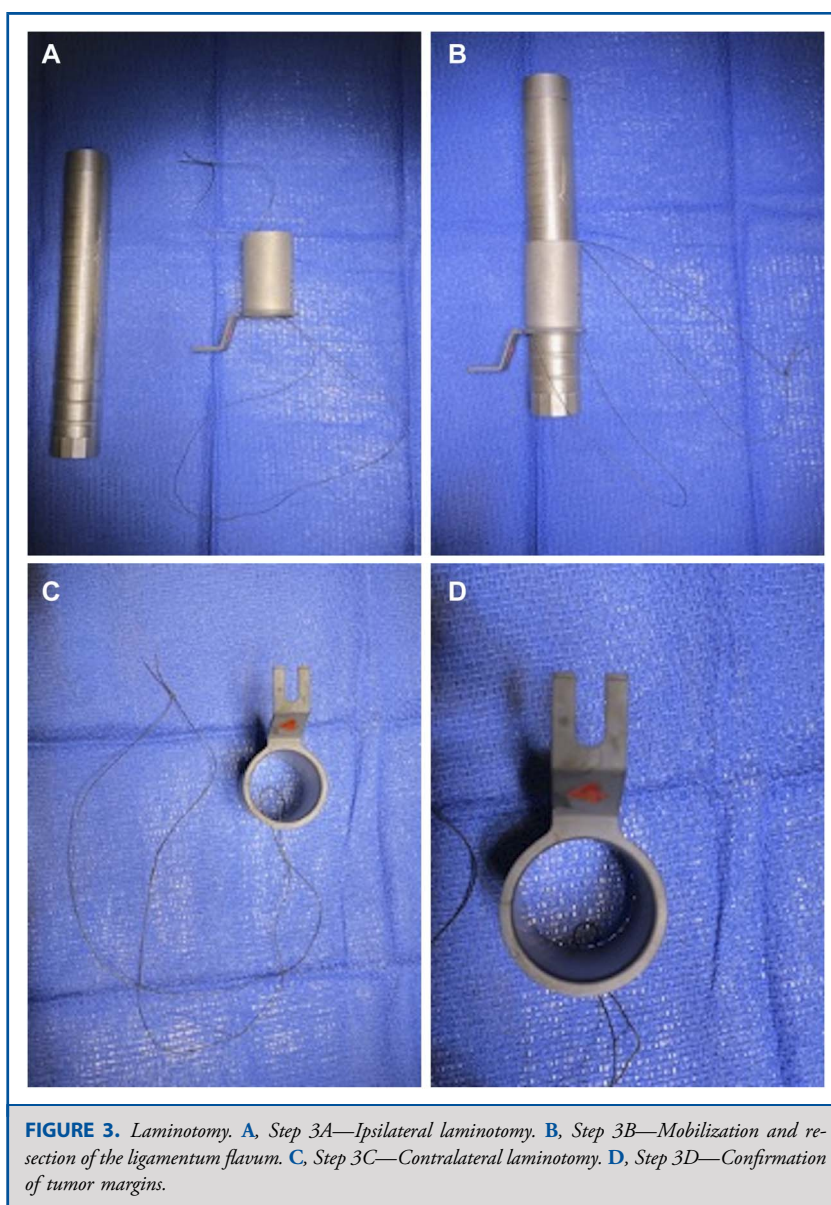


FIGURE 3. Laminotomy. **A**, Step 3A—Ipsilateral laminotomy. **B**, Step 3B—Mobilization and resection of the ligamentum flavum. **C**, Step 3C—Contralateral laminotomy. **D**, Step 3D—Confirmation of tumor margins.

selected (typically 4 cm or shorter). The placement of the tube is optimized using IONL. The microscope is brought in, and the target lamina is exposed with monopolar electrocautery (Figure 2B).

Laminotomy and Ligamentum Flavum Resection

A ULBD is performed in the manner previously detailed, including ligamentum removal and dural exposure.²⁵ (Figure 3) IONL can be used to ensure that the tumor margins fall within the tubular field of view.

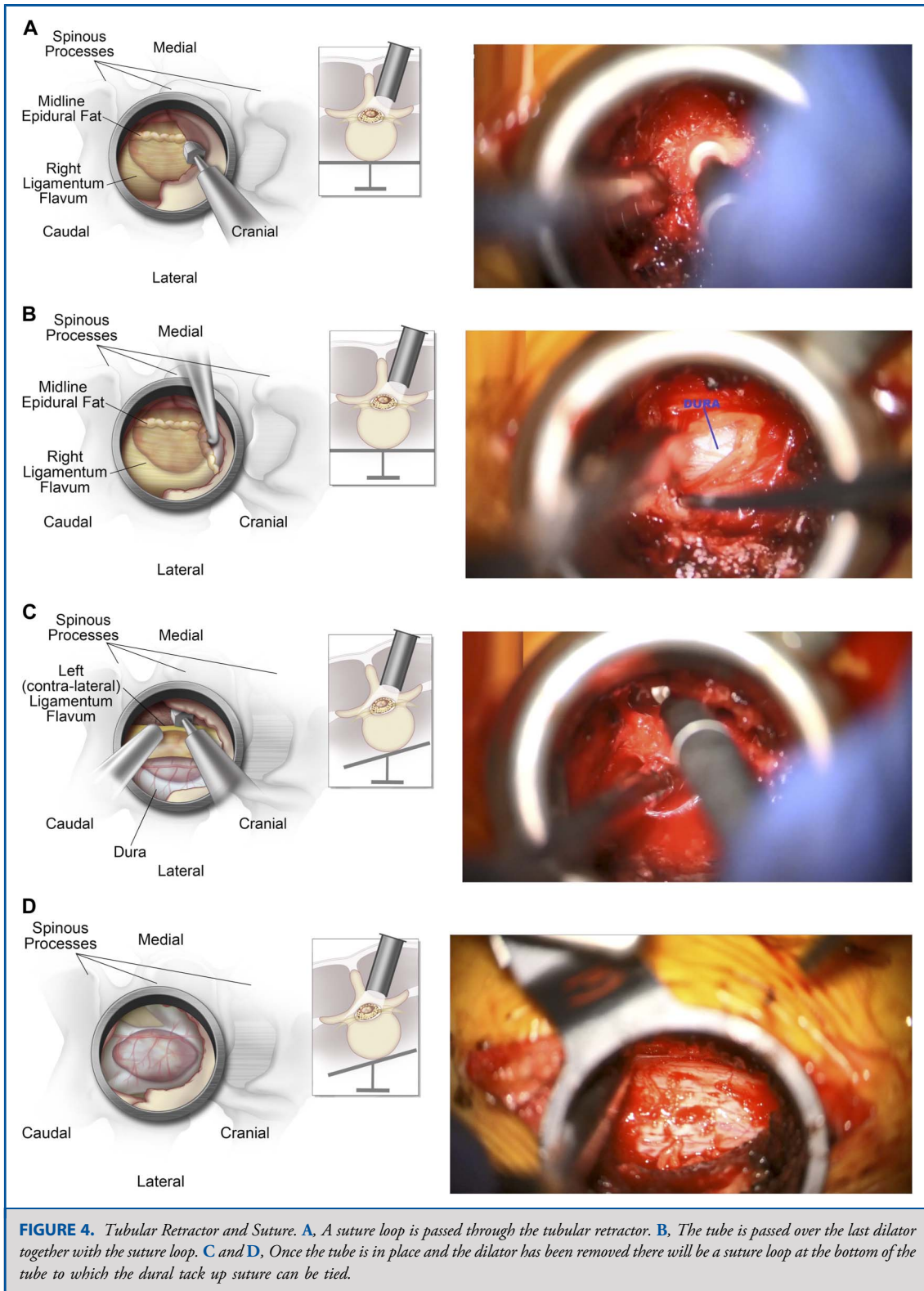
Dural Opening

ION is used to confirm the location, size, and orientation of the tumor. To maximize the dural opening, we developed a technique that allows placement of suture loops at the bottom of the tubular retractor to

which the dural tack-up sutures are later tied (Figure 4). To achieve this, the tubular retractor is passed over the last dilator with a tie that passes through the tube. This results in a suture loop at the bottom of the tube. The dura is linearly opened with an 11-blade. Next, 4-0 Nurodon TF-5 sutures are placed in the opposing dural edges and then tied to the suture loops at the periphery of the tube. The tails of the suture loops are then pulled up from outside the tube, spreading the dural edges out laterally instead of vertically, and facilitating a larger diameter intradural working space (Figure 5).

Tumor Resection

Careful microsurgical exploration is performed using microscissors and Rhoton dissectors until the mass is identified within the neural elements (Figure 6A). Identification and stimulation of neural elements



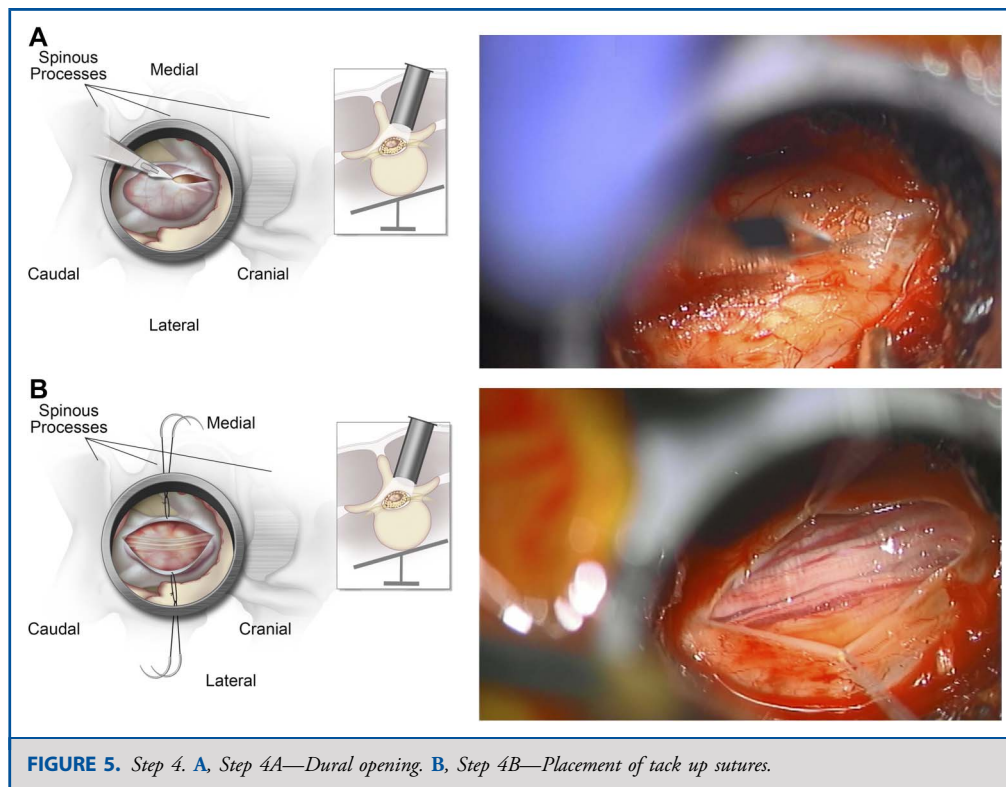


FIGURE 5. Step 4. **A**, Step 4A—Dural opening. **B**, Step 4B—Placement of tack up sutures.

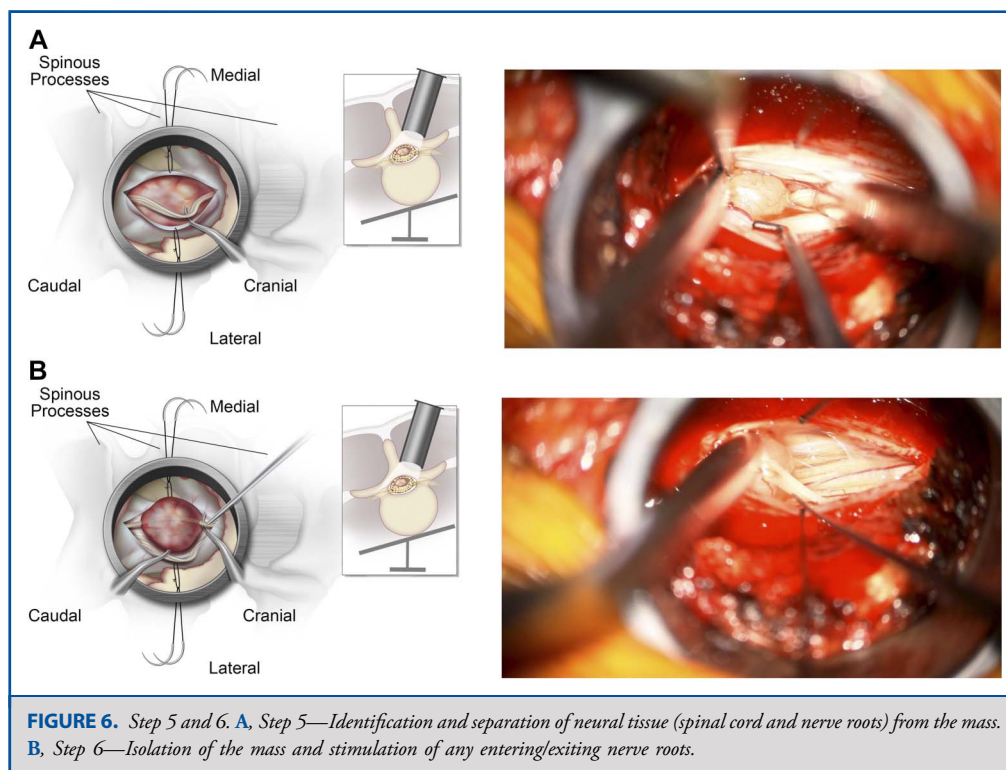
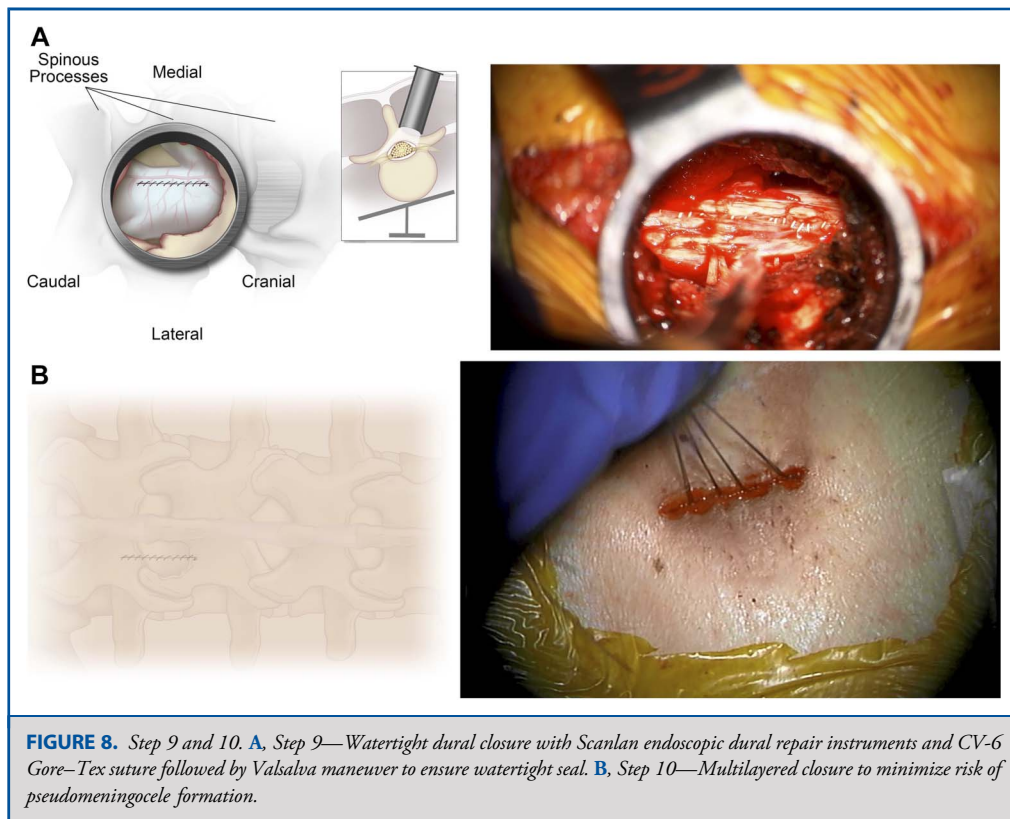
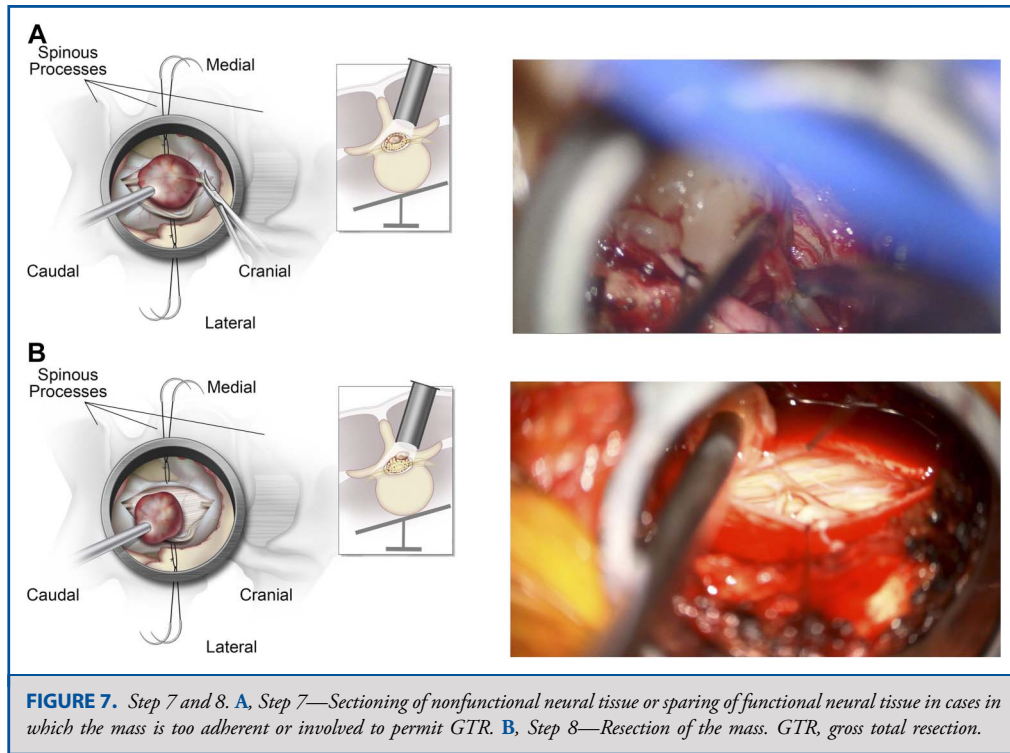


FIGURE 6. Step 5 and 6. **A**, Step 5—Identification and separation of neural tissue (spinal cord and nerve roots) from the mass. **B**, Step 6—Isolation of the mass and stimulation of any entering/exiting nerve roots.



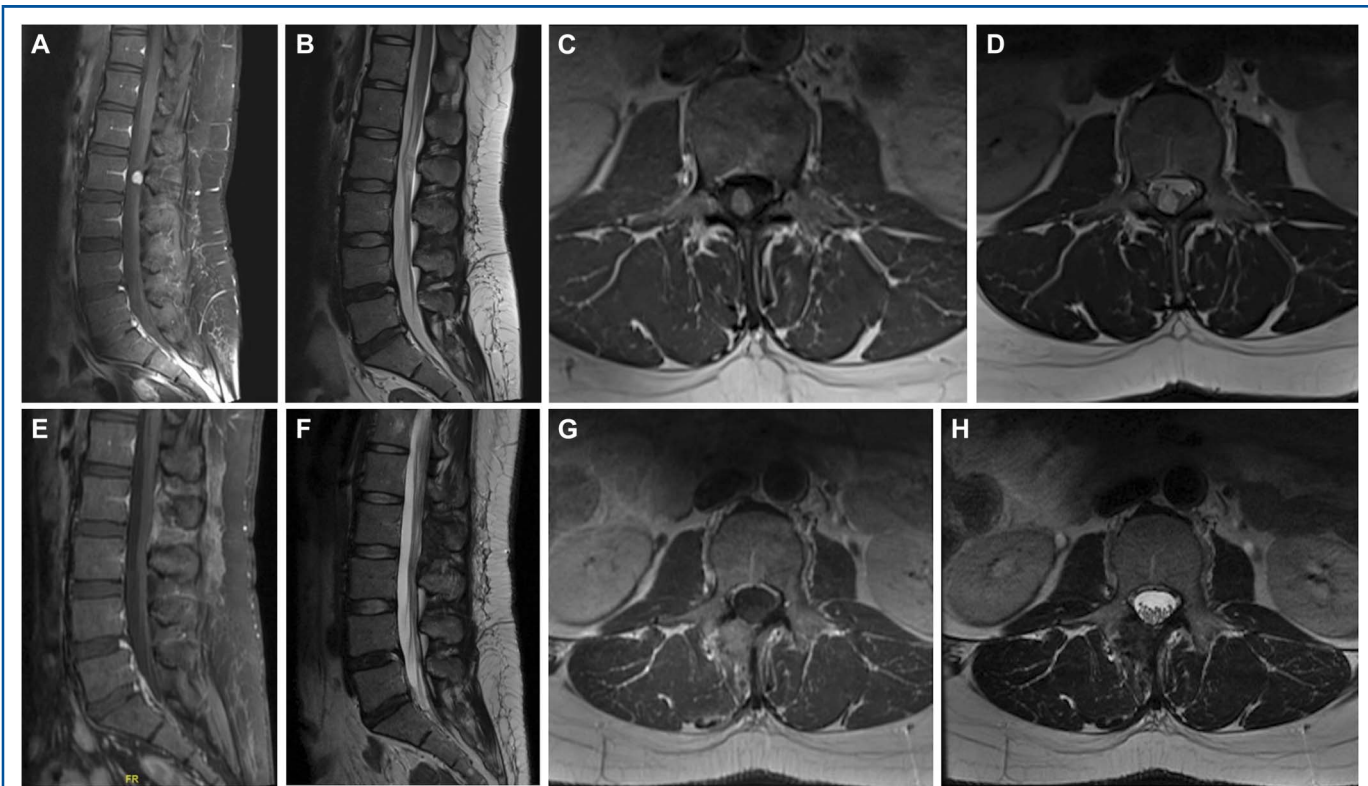


FIGURE 9. Preoperative and postoperative imaging. **A**, Preoperative sagittal T1 postcontrast MRI. **B**, Preoperative sagittal T2 MRI. **C**, Preoperative axial T1 postcontrast MRI. **D**, Preoperative axial T2 MRI. **E**, Postoperative sagittal T1 postcontrast MRI. **F**, Postoperative sagittal T2 MRI. **G**, Postoperative axial T1 postcontrast MRI. **H**, Postoperative axial T2 MRI (Note the relative absence of postoperative changes that result from a tubular approach).

(Figure 6B), sectioning of nonfunctional neural tissue or sparing of functional neural tissue (Figure 7A), and resection of the mass (Figure 7B) proceed in an identical fashion to open surgery.

Dural Closure

After delivery of the mass, hemostasis is achieved and any blood within the intradural space is irrigated out. We have previously described our approach to dural closure using the Scanlan dural closure kit and a 4-0 Nurodon TF-5 “fishhook.”²⁵ (Figure 8A; Video) Upon completion, a Valsalva maneuver is performed to confirm watertight closure and then fibrin sealant is applied over the suture line.

The tubular retractor is removed slowly under the microscope, taking care to cauterize any muscular bleeders. A drain is not typically used. The fascia and subcutaneous tissue are closed with absorbable sutures and the skin approximated using a running subcuticular suture (Figure 8B).

Postoperative Course

Postoperatively, patients are kept on overnight bedrest and monitored for signs of cerebrospinal fluid (CSF) leak before discharge for 24 to 48 hours.

Case Example

Presentation

A 35-year-old woman presented with low back pain that radiated into her legs, particularly at night. MRI demonstrated a 4-mm intradural mass at L2

consistent in appearance with a schwannoma (Figure 9A-9D). Given the severity and progressive nature of her symptoms, she was offered surgical resection. An MISS approach was selected because of the favorable size and position of the lesion, and the premium this young active patient placed on a rapid return to work and minimizing long-term biomechanical instability.

Surgical Description

The patient was positioned prone, a reference array was placed in the iliac crest, and an iCT was performed. IONL was used to guide incision planning and placement of the tubular retractor. As the tumor was eccentric to the left, a 26-mm tubular retractor was placed over the inferior lip of the left L2 lamina. An MIS tubular laminectomy was performed, exposing the dura. The margins of the tumor were mapped out using IONL, and the dura was opened using a bayoneted 11-blade knife and then tacked up bilaterally. The arachnoid was opened, and the nerve roots exposed and gently retracted to identify the tumor. In this case, a grayish mass was found adherent to multiple nerve roots which were carefully dissected off. Direct stimulation of the involved nerve root elicited no response, so it was cut and removed along with the tumor to achieve GTR. Intraoperative neuromonitoring, including somatosensory-evoked potentials and motor-evoked potentials were stable throughout.

Using a Scanlan endoscopic dural repair set, we closed the dura in a running fashion with a 4-0 Nurodon TF-5 “fishhook.” A Valsalva maneuver was performed to ensure watertight seal, and the rest of the closure was completed.

TABLE 2. Patient Data

Pt #	Age	Sex	Intradural and/or extradural	Level	Tumor diameter (mm)	Size of TR (mm)	Pathology	OR time	Anest. time	BL (cc)	LoS (d)	Follow-up time (mo)	Dural closure	Suture material
1	46	M	Intradural	S1-2	8 × 8 × 4	26	Schwannoma	3:14	5:08	15	2	1	Watertight	Gore-Tex
2	42	M	Intradural	L1-2	8 × 7 × 7	26	Schwannoma	2:45	4:22	10	1	1	Watertight	Gore-Tex
3	65	M	Intradural	L2-3	8 × 8 × 8	24	Paraganglioma	2:21	3:41	150	2	13	Watertight	Gore-Tex
4	55	F	Intradural	S1-2	16 × 14 × 13	24	Schwannoma	2:47	4:05	50	2	9	Watertight	Gore-Tex
5	58	M	Intradural and extradural	L1-2	11 × 8 × 8	24	Schwannoma	5:08	6:42	500	2	18	Watertight	Gore-Tex
6	34	F	Intradural	L2-3	12 × 10 × 8	24	Myxopapillary ependymoma	2:29	3:14	10	2	15	Watertight	Gore-Tex
7	60	F	Intradural	T12-L1	28 × 18 × 18	24	Meningioma	3:05	4:30	50	2	40	Watertight	Nurolon
8	78	M	Intradural	L3-L4	5 × 5 × 5	24 and 18	Schwannoma	2:44	4:07	25	3	41	Watertight	Nurolon
9	41	M	Intradural	L1	8 × 5 × 5	24	Schwannoma	3:38	4:56	500	3	4	Watertight	Nurolon
10	32	M	Intradural	T12-L1	19 × 15 × 15	24	Schwannoma	3:06	4:41	10	2	1	Watertight	Nurolon
11	64	M	Intradural and extradural	L3-4	15 × 10 × 9	21	Schwannoma	3:35	5:41	220	5	45	Watertight	Nurolon
12	37	M	Intradural	L2-3	22 × 14 × 13	24	Myxopapillary ependymoma	4:17	5:58	25	1	36	Watertight	Nurolon
13	62	F	Intradural	L3-4	14 × 12 × 11	25	Schwannoma	2:17	3:37	20	3	4	Watertight	Nurolon
14	35	M	Extradural	L4-5	34 × 19 × 18	25	Schwannoma	3:14	4:52	100	2	21	N/A	N/A
15	46	M	Intradural	L1-2	16 × 15 × 13	24	Schwannoma	2:42	4:12	50	1	29	Watertight	Nurolon
16	46	F	Extradural	L4-5	29 × 18 × 16	21	Schwannoma	3:06	4:40	200	5	2	N/A	N/A
17	70	M	Intradural	L2-3	13 × 10 × 9	24	Schwannoma	2:43	4:31	200	1	53	Watertight	Nurolon
18	66	F	Intradural	L2-3	12 × 9 × 7	26	Schwannoma	3:42	5:50	10	3	66	Watertight	Nurolon
19	82	M	Intradural	L4-5	8 × 7 × 6	26	Paraganglioma	2:48	4:23	75	1	3	Watertight	Nurolon
20	60	M	Intradural	L2-3	10 × 6 × 4	26	Myxopapillary ependymoma	3:44	5:18	300	3	1	Watertight	Nurolon

The patient did well and was discharged after 2 days. Follow-up imaging at 6 months demonstrated radiological GTR and no sign of pseudomeningocele (Figure 9E-9H).

RESULTS

Collected data are presented in Table 2. The mean age of patients was 54 years (range: 32-82 years). There was a male predominance ($n = 14$) in our cohort. The majority ($n = 16$) of our patients had IDEM tumors; however, 2 patients had tumors with both intradural and extradural components, and 2 patients had strictly extradural tumors. Most tumors ($n = 16$) were lumbar. The mean tumor size was 14.8 mm, which represents the average of the maximum dimension of each tumor. The average blood loss was 126 cc (range 10-500 cc), and the operation duration was 190 minutes (range 137-308 minutes). One patient simultaneously underwent ULBD at a remote level for coincidental central canal stenosis (5%). No intraoperative complications were observed. Based on final pathological reports, 14 patients were diagnosed with schwannoma (70%), 3 with myxopapillary ependymoma (15%), 2 with paraganglioma (10%), and 1 with

meningioma (5%). The average hospital length of stay was 2.3 days (range 1-5 days). No patients had recurrence of tumor during follow-up period mean for which was 20.2 months (range 1-66 months).

DISCUSSION

Here, we present the first case series to demonstrate a workflow integrating microsurgical tubular resection and total navigated localization for the treatment of IDEM tumors. We feel that the current study is an important step in the development of a systematic approach to the appropriate application of MISS techniques to intradural pathology that is practical for widespread adoption.

Our data demonstrate that through careful patient selection and the utilization of an integrated MIS workflow, excellent results can be achieved in the resection of IDEM tumors. GTR was achieved in all 20 patients treated in our cohort. This result aligns with the GTR rates reported in the literature for both open (74%-92%) as well as MIS and mini-open (66.6%-100%) approaches.^{1,4,6,17,42,43} Our result was durable with no cases of

recurrent or residual tumor identified on a mean follow-up of 20.2 months.

Our findings also demonstrate a mean hospital length of stay of 2.3 days. This aligns with the previously published length of stay values for MIS vs open IDEM tumor resection (3.9 and 6.1 days, respectively).⁶ Decreased postoperative pain and an extremely low rate of perioperative complications may be presumed to drive this reduction in hospital duration. In our study, no patient required a lumbar drain or developed a symptomatic cerebrospinal fluid leak or postoperative pseudomeningocele. This result is also reflected in the literature in which reported cerebrospinal fluid leak rates after IDEM tumor resection range from 0% to 5% for MIS surgery to 0% to 9% for open surgery.^{6,7,16-18,36,44,45}

We acknowledge that radiation exposure using iCT poses a dilemma. With fluoroscopy, only 2 to 3 shots for incision localization and confirmation of docking position/level are required. The utility of iCT and navigation comes with being able to precisely plan the laminectomy cranially-caudally and confirmation of contralateral decompression to ensure adequate dural exposure. Since iCT and navigation are used for these portions of the case, we also use it for retractor docking for efficiency. The iCT used in our study has radiation reduction modes, which reduce radiation to the patient by 50% to 90%.^{46,47} Additional operative and anesthesia time is also a concern with iCT because of setup time and image acquisition, although studies of lumbar fusion have demonstrated negligible differences between these 2 modalities.⁴⁸⁻⁵⁰ Nonetheless, we acknowledge that with noninstrumented, one-level cases without deformity, this additional time must be considered. Having staff familiar with the iCT workflow^{40,49} is paramount to maximizing efficiency.

Limitations

Although our data suggest that the navigated localization and microsurgical tubular resection of IDEM tumors is safe and effective for appropriately selected patients, our study does have limitations. First, we do not compare our results to a control group. Although we describe excellent outcomes for our patient cohort, we have insufficient data to conclude these patients would not have had an equivalent outcome with an open technique. Second, for slow-growing tumors, our follow-up period may not capture a minority of patients who may develop delayed recurrence from microscopic residual tumor. Third, patient-reported outcomes were not compared. Nonetheless, this study represents an important step in describing a systematic approach applying MISS principles and technology for the treatment of IDEM tumors.

CONCLUSION

Our findings suggest that the MIS approach can be safely and effectively used in the treatment of carefully selected patients with IDEM tumors through the adoption of a surgical workflow that integrates intraoperative navigated localization with minimally invasive tubular retractors and the operative microscope.

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Disclosures

Dr McGrath has equity interest in EigenHealth, Inc. Dr Hartl is a consultant for Ulrich, Brainlab, and DePuy-Synthes; has royalties from Zimmer; and has financial relationships with RealSpine and 3D Bio. Dr Sommer received an educational grant for his fellowship from Brainlab and has speaking fees from Baxter.

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VIDEO. Technique for dural closure through a tubular retractor.
